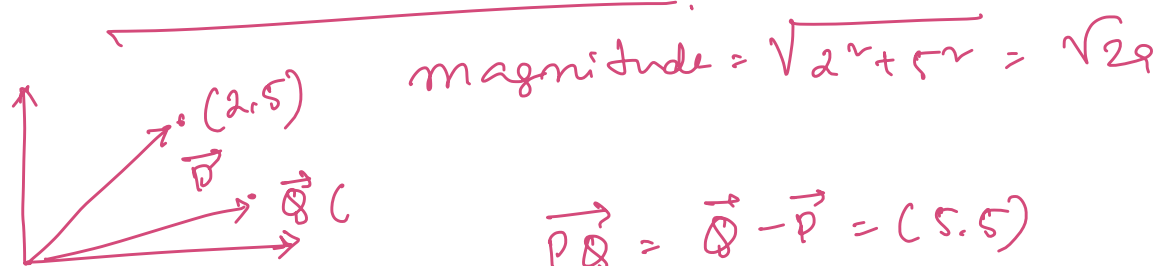


Support Vector Machine

Coordinate Geometry

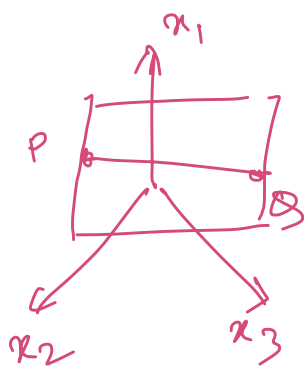


$$\vec{PQ} = \vec{Q} - \vec{P} = (5, 5)$$

$$\vec{Q} (7, 10)$$

$$\vec{P} (2, 5)$$

$$\text{Angle } \tan \theta = y/x \Rightarrow \theta = \tan^{-1}(y/x)$$



$$f(x) = w_1 x_1 + w_2 x_2 + w_3 x_3 + w_0 = 0$$

$$w^T x + w_0 = 0$$

$$w = \{w_1, w_2, \dots, w_3\}$$

$$x = \{x_1, x_2, x_3\}$$

Normal vector of $f(x)$

$$\vec{w} = [w_1, w_2, w_3] \Rightarrow \nabla f(x)$$

show that

$$\vec{PQ} \cdot \vec{w} = 0$$

$$\vec{P} = w_1 p_1 + w_2 p_2 + w_3 p_3 + w_0 \Rightarrow w^T P = -w_0$$

$$\vec{Q} = w_1 q_1 + w_2 q_2 + w_3 q_3 + w_0$$

$$w^T Q = -w_0$$

$$w^T P - w^T Q = 0$$

$$\Rightarrow w^T (P - Q) = 0$$

$$\Rightarrow \underline{w^T \cdot \vec{PQ} = 0}$$

distance of a point x from the plane, $w^T x + w_0 = 0$

$$d = \frac{w^T x + w_0}{|w_0|} = \hat{\gamma} = \underline{\text{geometric distance}}$$

Functional distance $w^T x + w_0 = \hat{\gamma}$

